

assignment power_gravity BONUS!! ⚡

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:40pm

Quiz Instructions

no tutorial. but hints. Hide the hints first



Question 1 10 pts

How long will it take a 7.08 hp motor to lift a 250 kg beam directly upward at constant velocity from the ground to a height of 45.0 m? Assume frictional forces are negligible. (1 hp = 746 W)

hint: you can use power = work done/time.

work against gravity on beam= weight x h

(the work done by the thrust is used to change the PE). convert hp to W

☐

20.9 s

☐

39.7 s

☐

1.56×10^4 s

☐

2.18×10^4 s



Question 2 8.5 pts

A car needs to generate 75W in order to maintain a constant velocity of 27.3 m/s on a flat road. What is the magnitude of the total resistive force acting on the car (due to friction, air resistance, etc.)? (1 hp = 746 W)

hint: Its moving at a constant speed. So thrust=friction. . power = thrust x (average speed).

☐

2.05×10^3 N

☐

1.03×10^3 N

☐

2.75N



$2.87 \times 10^3 \text{ N}$



Question 3 8.5 pts

The force of Earth's gravity on a capsule in space will lessen as it moves farther away. If the capsule moves to twice its distance, the force toward Earth becomes

hint: The force of gravity follows an inverse square law as discussed in class (see my slideS)



none of the above



three-quarters.



half.



one-fourth.



Question 4 8.5 pts

Which task requires more work?

hint: what is the definition of work?



need more information



both require the same



lifting the 50-kg sack 2 meters



lifting the 25-kg sack 4 meters



Question 5 8.5 pts

The power expended doing 100 J of work in 50 s is

hint: power is work/time or force dot velocity



4 W.



50 W.



5,000 W.



1/2 W.



2 W.



Question 6 10 pts

As two objects moving toward each other due to gravity get closer, the acceleration of each

hint: The gravitational force between 2 masses is $G m M / r^2$. r is the distance between the 2 masses (M and m). G is a constant. If you consider one of those masses (m) then $F = ma$
What is a ? (acceleration due to gravity).



decreases.



remains constant.



increases.



Question 7 8.5 pts

One kilowatt-hour represents

hint: kilo is 1000 and 1 hour is seconds.



60,000 W.



60 N/s.



none of these



3.6 million joules.



1 N.



Question 8 12 pts

If you bounce up and down on a bathroom scale, what varies on the scale reading is



neither of these



both of these



the normal force.



mg .



Question 9 8.5 pts

A particle experiences a force given by $F(x) = \alpha - \beta x^3$. Find the potential field $U(x)$ the particle is in. (Assume that the zero of potential energy is located at $x = 0$.)

hint:

$F(x)$ is the restoring force. To put energy in the system, work has to be done against that force = - work done by the restoring force = - integral ($F_{\text{restoring}} dx$).

so by definition $\text{change}(U) = - \text{integral} (F_{\text{restoring}} dx)$.

$U(x) - U(0) = - \text{integral} (F_{\text{restoring}} dx)$. See slide for definition.

☐

$$U(x) = -\alpha x + (\beta/4) x^4$$

☐

$$U(x) = 3\beta x^2$$

☐

$$U(x) = \alpha x - (\beta/4)x^4$$

☐

$$U(x) = -3\beta x^2$$

⋮

Question 10 12 pts

What is the gravitational force acting on a 59-kg person due to another 59-kg person standing 2.0 m away? We can model each person as a small sphere. ($G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$)

hint: $F = G m M / r^2$. See slide.

☐

$$2.0 \times 10^{-9} \text{ N}$$

☐

$$1.2 \times 10^{-7} \text{ N}$$

☐

$$9.8 \times 10^{-10} \text{ N}$$

☐

$$5.8 \times 10^{-8} \text{ N}$$

☐

$$8.5 \times 10^3 \text{ N}$$

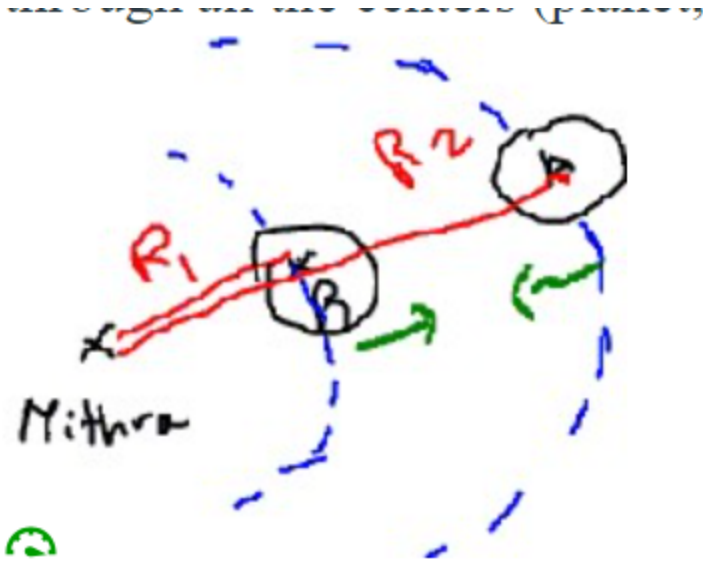
⋮

Question 11 8.5 pts

	Mass	Radius	Orbital radius	Orbital period
Moon A	4.0×10^{20} kg	unknown	2.0×10^8 m	4.0×10^6 s
Moon B	1.5×10^{20} kg	2.0×10^5 m	3.0×10^8 m	unknown

Mithra is an unknown planet that has two moons, A and B, in circular orbits around it. The table summarizes the hypothetical data about these moons. What is the magnitude of the *maximum* gravitational force that Moon A exerts on Moon B? ($G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$)

hint: Its actually easy. First draw the systems. When the force is max when the 2 Moons are aligned. There is one line going through all the centers (planet, A and B)



So distance between the moons is $R_2 - R_1 = r$ (difference in orbital radius)

$$F = G M_A M_B / r^2.$$

- ☐ $4.4 \times 10^{13} \text{ N}$
- ☐ $1.6 \times 10^{13} \text{ N}$
- ☐ $4.0 \times 10^{14} \text{ N}$
- ☐ $1.0 \times 10^{14} \text{ N}$

 $2.0 \times 10^{14} \text{ N}$ 

Question 12 12 pts

As a 70-kg person stands at the seashore gazing at the tides (which are caused by the Moon), how large is the gravitational force on that person due to the Moon? The mass of the Moon is $7.35 \times 10^{22} \text{ kg}$, the distance to the Moon is $3.82 \times 10^8 \text{ m}$, and $G = 6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$.
hint: Use $F = G m M / (r^2)$



0.0024 N



0.24 N



0.00024 N



0.024 N

Not saved

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